

Kinetic Modelling of Ordered Fluids

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Mathematical Institute, University of Oxford

Supervisors: P. E. Farrell, J. A. Carrillo

DPhil Viva Voce Examination



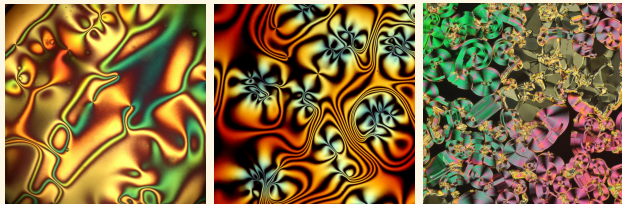
Mathematical
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Oxford
Mathematics



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ORDERED FLUIDS



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- ▶ Liquid crystals, shown here in nematic and smectic phases.
- ▶ Ferrofluids, colloidal suspensions of nanoscale ferromagnetic or ferrimagnetic particles.
- ▶ Gas-saturated magma melts, and other fluids carrying bubbles that do not diffuse.

OUTLINE

- | | |
|---|---------------|
| 1. Order parameter manifold | Chapters 2–3 |
| 2. The microscopic world | Chapter 2 |
| 3. BBGKY hierarchy | Chapters 4–5 |
| 4. Vlasov–Boltzmann equation | Chapter 6 |
| 5. Direct simulation Monte Carlo | Chapters 7–8 |
| 6. Gradient-flow structure of the marginal dynamics | Chapters 9–11 |

 *Multiscale Model. Simul.*, in press, arXiv:2508.10744 (J. A. Carrillo, P. E. Farrell, A. Medaglia, U. Z.), *Multiscale Model. Simul.* 2024, 22, 1585–1607 (P. E. Farrell, G. Russo, U. Z.), arXiv 2026, 2609.31072 (J. A. Carrillo, P. E. Farrell, A. Medaglia, U. Z.), *In preparation* (J. A. Carrillo, U. Z.).

Chapter 11 is being written up with J. A. Carrillo as *One-dimensional gradient-flow structure of the marginal dynamics of a kinetic model of ordered fluids*.

ORDER PARAMETER MANIFOLD

1

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Order Parameter Manifold

We say that the tuple $(\mathcal{M}, \mathcal{A})$ is an order parameter manifold if $\mathcal{M}$ is a smooth compact manifold with a fixed parametrisation, and $\mathcal{A}$ is a Lie group action of $SO(d)$ on $\mathcal{M}$.



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(G. Capriz),
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A field $\nu : \mathbb{E}^d \rightarrow \mathcal{M}$ is an order parameter field if, under a change of frame $\mathbf{x} \mapsto \underline{\underline{Q}}\mathbf{x} + \underline{\underline{c}}$, the field ν^* describing the same microstructure is

$$\nu^*(\underline{\underline{Q}}\mathbf{x} + \underline{\underline{c}}) = \mathcal{A}(\underline{\underline{Q}}, \nu(\mathbf{x})), \quad \forall \mathbf{x} \in \mathbb{E}^d.$$



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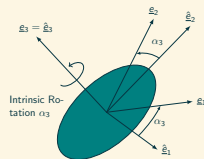
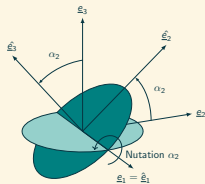
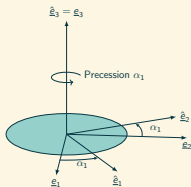
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- ▶ Which manifold $\mathcal{M}$ describes the order of a given fluid?
- ▶ How do rotations act on $\mathcal{M}$?

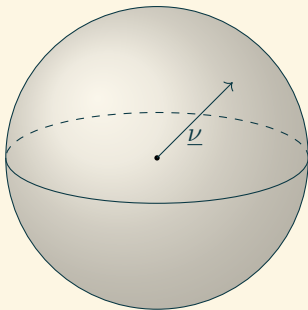
AN EXAMPLE: NEMATIC LIQUID CRYSTALS



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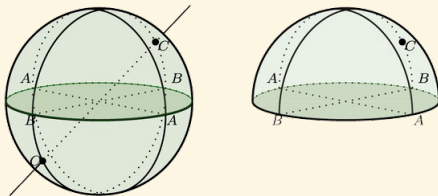
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- ▶ The state of a rod-like (calamitic) molecule can be described by its Euler angles θ, ϕ, ψ ,
- ▶ or by a director $\underline{v} \in \mathbb{S}^2$.
- ▶ If head and tail are indistinguishable, the director lives in $\mathbb{RP}^2$.

EMBEDDING RESULTS

Embedding theorems

- ▶ Every compact orientable surface embeds in $\mathbb{R}^3$.



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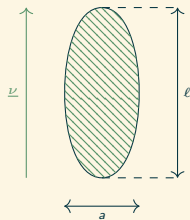
\$\$ and a point of $\mathbb{RP}^2$ as a vector of $\mathbb{R}^4$.

THE MICROSCOPIC WORLD

2

LAGRANGIAN MECHANICS OF THE CONSTITUENTS

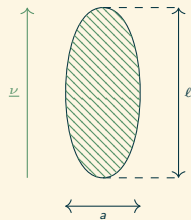
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$$\mathcal{L}_i := \frac{1}{2} m (\dot{\mathbf{x}}_i \cdot \dot{\mathbf{x}}_i) + \frac{1}{2} \dot{\nu}_i \cdot \underline{B}_i(\nu_i) \dot{\nu}_i.$$



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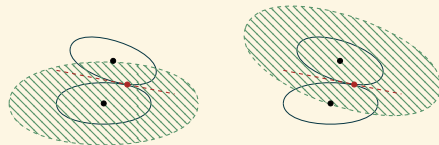
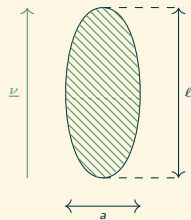
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Two constituents interact through a potential $\mathcal{W}(\mathbf{x}_i - \mathbf{x}_j, \nu_i, \nu_j)$,

$$\mathcal{L}_{i,j} = \mathcal{L}_i(\mathbf{x}_i, \Xi_i) + \mathcal{L}_j(\mathbf{x}_j, \Xi_j) + \mathcal{W}(\mathbf{x}_i - \mathbf{x}_j, \nu_i, \nu_j),$$

where $\Xi_i := (\underline{v}_i, \nu_i, \dot{\nu}_i)$.



NOETHER'S THEOREM: SYMMETRIES AND CONSERVATION LAWS

Noether's theorem

If a Lagrangian $\mathcal{L}$ is invariant under a group action with infinitesimal generators G then

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_{1,2}} \cdot G \right) = 0, \quad q_{1,2} = (\mathbf{x}_1, \mathbf{x}_2, \nu_1, \nu_2).$$

Each symmetry of the system gives a conserved quantity.



Analytical Mechanics: An Introduction, (A. Fasano, S. Marmi).

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- ▶ $\mathcal{L}$ is invariant under translations, so linear momentum is conserved.
- ▶ $\mathcal{L}$ does not depend on time and its kinetic energy is a homogeneous quadratic form in the generalised velocities, so energy is conserved.

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Infinitesimal Generator of $\mathcal{A}$

For fixed $\nu \in \mathcal{M}$, the orbit map

$$\mathcal{A}_\nu : \mathrm{SO}(3) \rightarrow \mathrm{SO}(3)\nu, \quad \underline{Q} \mapsto \mathcal{A}(\underline{Q}, \nu),$$

is differentiable at the identity. Its differential there is the infinitesimal generator $A_\nu : \mathfrak{so}(3) \rightarrow T_\nu \mathcal{M}$, which we also read as a map $\mathbb{R}^3 \rightarrow T_\nu \mathcal{M}$ through $\mathbb{R}^3 \cong \mathfrak{so}(3)$.



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If $\mathcal{L}$ is frame indifferent, that is invariant under $\text{SO}(3)$, the generators are

$$G = (\underline{r} \times \mathbf{x}_1, \underline{r} \times \mathbf{x}_2, A_{\nu_1} \underline{r}, A_{\nu_2} \underline{r}),$$

with $\underline{r}$ the rotation axis, so angular momentum is conserved.

AN EXAMPLE: ANGULAR MOMENTUM NEMATIC LIQUID CRYSTALS

For segment-like molecules $\underline{\underline{B}}(\underline{\nu}) = I$, the identity, and Noether's theorem conserves

$$\underline{\eta} = \mathbf{x}_1 \times \underline{p}_1 + \underline{\nu}_1 \times \dot{\underline{\nu}}_1 + \mathbf{x}_2 \times \underline{p}_2 + \underline{\nu}_2 \times \dot{\underline{\nu}}_2.$$

A segment with angular velocity $\underline{\omega}_i$ has $\dot{\underline{\nu}}_i = \underline{\omega}_i \times \underline{\nu}_i$, so by the triple product

$$\underline{\nu}_i \times \dot{\underline{\nu}}_i = \underline{\nu}_i \times (\underline{\omega}_i \times \underline{\nu}_i) = (\underline{\nu}_i \cdot \underline{\nu}_i) \underline{\omega}_i - (\underline{\nu}_i \cdot \underline{\omega}_i) \underline{\nu}_i = \underline{\omega}_i - (\underline{\nu}_i \cdot \underline{\omega}_i) \underline{\nu}_i = \mathbb{I}_i \underline{\omega}_i,$$

where $\mathbb{I}_i := I - \underline{\nu}_i \otimes \underline{\nu}_i$ is the inertia tensor of the segment. This is the classical angular momentum,

$$\mathbf{x}_1 \times \underline{p}_1 + \mathbb{I}_1 \underline{\omega}_1 + \mathbf{x}_2 \times \underline{p}_2 + \mathbb{I}_2 \underline{\omega}_2,$$

BBGKY HIERARCHY

3

HAMILTONIAN MECHANICS OF THE CONSTITUENTS

In the Hamiltonian picture the conjugate momenta are

$$\underline{p}_i := \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{x}}_i} = m \dot{\mathbf{x}}_i, \quad \underline{s}_i := \frac{\partial \mathcal{L}}{\partial \dot{\nu}_i} = \underline{\underline{B}}(\nu) \dot{\nu}_i.$$

and N constituents interacting in pairs have the Hamiltonian

$$\mathcal{H} := \sum_{i=1}^N \frac{1}{2m} \underline{p}_i \cdot \underline{p}_i + \frac{1}{2} \underline{s}_i \cdot \underline{\underline{B}}(\nu)^{-1} \underline{s}_i + \sum_{1 \leq i < j \leq N} \mathcal{W}(\mathbf{x}_i - \mathbf{x}_j, \nu_i, \nu_j).$$

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The Legendre transform of $\mathcal{L}$ is well defined as long as $\underline{\underline{B}}(\nu)$ is symmetric and positive definite for every $\nu \in \mathcal{M}$.

LIOUVILLE'S THEOREM ON THE ORDER PARAMETER MANIFOLD

The state $(\mathbf{x}, \nu, \underline{p}, \underline{\varsigma})$ of a constituent is a point of the cotangent bundle $T^*(\mathbb{E}^d \times \mathcal{M})$, since $\underline{\varsigma} = \underline{\underline{B}}(\nu)\underline{\dot{\nu}}$ is a covector at ν .



Introduction to Mechanics and Symmetry, (J. E. Marsden, T. S. Ratiu),
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- ▶ Every cotangent bundle carries a canonical symplectic form, locally $\sum_i dq_i \wedge d\pi_i$ with $q = (\mathbf{x}, \nu)$, $\pi = (\underline{p}, \underline{\varsigma})$, and with it the Poisson bracket

$$\{F, \mathcal{H}\} = \sum_i \nabla_{q_i} F \cdot \nabla_{\pi_i} \mathcal{H} - \nabla_{\pi_i} F \cdot \nabla_{q_i} \mathcal{H}.$$



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$$\partial_t f_N + \{f_N, \mathcal{H}\} = 0.$$



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Only the smoothness of $\mathcal{M}$ is used. It need not be a vector space or even-dimensional, as $\mathbb{S}^1$, $\mathbb{RP}^1$ and $[0, 1]$ show.

In the canonical variables $(\nu, \underline{\varsigma})$ the Liouville volume is $d\nu d\underline{\varsigma}$. In $(\nu, \underline{\dot{\nu}})$ it would carry a factor $\det \underline{\underline{B}}(\nu)$.

BOGOLIUBOV–BORN–GREEN–KIRKWOOD–YVON HIERARCHY



An Introduction to the Theory of the Boltzmann Equation, (S. Harris),
Statistical Physics of Particles, (M. Kardar),
Statistical Mechanics, 2nd Edition (K. Huang).

Integrating Liouville's equation over $\Gamma_{s+1}, \dots, \Gamma_N$, the normalised s -particle distribution function f_s satisfies

$$\begin{aligned} \frac{\partial f_s}{\partial t} + \{f_s, \mathcal{H}_s\} = & \int \sum_{i=1}^s \frac{\partial f_{s+1}}{\partial \underline{p}_i} \cdot \frac{\partial \mathcal{W}(\mathbf{x}_i - \mathbf{x}_{s+1}, \nu_i, \nu_{s+1})}{\partial \mathbf{x}_i} d\Gamma_{s+1} \\ & + \int \sum_{i=1}^s \frac{\partial f_{s+1}}{\partial \underline{\zeta}_i} \cdot \frac{\partial \mathcal{W}(\mathbf{x}_i - \mathbf{x}_{s+1}, \nu_i, \nu_{s+1})}{\partial \nu_i} d\Gamma_{s+1}, \end{aligned}$$

where

$$\mathcal{H}_s = \sum_{i=1}^s \left(\frac{|\underline{p}_i|^2}{2m} + \frac{1}{2} \underline{\zeta}_i \cdot \underline{B}(\nu_i)^{-1} \underline{\zeta}_i \right) + \sum_{1 \leq i < j \leq s} \mathcal{W}(\mathbf{x}_i - \mathbf{x}_j, \nu_i, \nu_j).$$

BOGOLIUBOV–BORN–GREEN–KIRKWOOD–YVON HIERARCHY

Without three-body interactions, the first two equations of the hierarchy are

$$\begin{aligned} \frac{\partial f_1}{\partial t} + \frac{\underline{p}_1}{m} \cdot \frac{\partial f_1}{\partial \mathbf{x}_1} + \underline{\underline{B}}(\nu_1)^{-1} \underline{\zeta}_1 \frac{\partial f_1}{\partial \nu_1} = \\ + \int \frac{\partial \mathcal{W}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2)}{\partial \mathbf{x}_1} \left(\frac{\partial f_2}{\partial \underline{p}_1} - \frac{\partial f_2}{\partial \underline{p}_2} \right) \\ + \int \frac{\partial \mathcal{W}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2)}{\partial \nu_1} \left(\frac{\partial f_2}{\partial \underline{\zeta}_1} - \frac{\partial f_2}{\partial \underline{\zeta}_2} \right) \end{aligned}$$

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WHY THE THREE-BODY TERM DROPS: THE BOLTZMANN-GRAD LIMIT

Let d be the interaction range, c the typical speed, n the density and L a macroscopic length. The time scales are

$$\underbrace{\tau_c = \frac{d}{c}}_{\text{a collision}}, \quad \underbrace{\tau = \frac{1}{nd^2c}}_{\text{between collisions}}, \quad \underbrace{T = \frac{L}{c}}_{\text{macroscopic}}.$$

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Take $\mathcal{W} \sim c^2$, vanishing beyond d . Since $\int f_3 d\Gamma_3 \sim nf_2$,

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WHY THE THREE-BODY TERM DROPS: THE BOLTZMANN–GRAD LIMIT

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Let $n \rightarrow \infty$ and $d \rightarrow 0$ with nd^2 fixed. The mean free path stays fixed while $nd^3 \rightarrow 0$, so

$$\frac{\text{three-body}}{\text{two-body}} \sim \frac{\tau_c}{\tau} = nd^3 \rightarrow 0.$$

Collisions become binary and instantaneous, and the right-hand side of the f_2 equation can be dropped.

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Collisions become binary and instantaneous, and the right-hand side of the f_2 equation can be dropped.

In $\mathbb{R}^D$ the cross-section scales as d^{D-1} , and the scaling reads nd^{D-1} fixed, $nd^D \rightarrow 0$.

FAST AND SLOW TERMS IN THE TWO-PARTICLE EQUATION

With $\mathbf{x} = \mathbf{x}_2 - \mathbf{x}_1$ and $\mathbf{X} = \frac{1}{2}(\mathbf{x}_2 + \mathbf{x}_1)$ the f_2 equation reads

$$\begin{aligned} \frac{\partial f_2}{\partial t} + \frac{1}{2} \frac{\underline{p}_2 + \underline{p}_1}{m} \cdot \frac{\partial f_2}{\partial \mathbf{X}} + \underline{\underline{B}}(\nu_1)^{-1} \underline{\underline{\varsigma}}_1 \cdot \frac{\partial f_2}{\partial \nu_1} + \underline{\underline{B}}(\nu_2)^{-1} \underline{\underline{\varsigma}}_2 \cdot \frac{\partial f_2}{\partial \nu_2} + \boxed{\frac{\underline{p}_2 - \underline{p}_1}{m} \cdot \frac{\partial f_2}{\partial \mathbf{x}}} \\ - \boxed{\frac{\partial \mathcal{W}}{\partial \mathbf{x}_1} \cdot \left(\frac{\partial f_2}{\partial \underline{p}_1} - \frac{\partial f_2}{\partial \underline{p}_2} \right)} - \frac{\partial \mathcal{W}}{\partial \nu_1} \cdot \left(\frac{\partial f_2}{\partial \underline{\varsigma}_1} - \frac{\partial f_2}{\partial \underline{\varsigma}_2} \right) = 0. \end{aligned}$$

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A collision leaves $\dot{\mathbf{X}}$ unchanged,

$\frac{d}{dt} \dot{\mathbf{X}} = -\frac{1}{2}(\nabla_{\mathbf{x}_1} + \nabla_{\mathbf{x}_2})\mathcal{W} = 0$, so f_2 varies in $\mathbf{X}$ on the mean free path, $\dot{\mathbf{X}} \cdot \nabla_{\mathbf{X}} f_2 \sim f_2/\tau$.

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On τ_c the boxed terms balance. This needs $\mathcal{W}$ to depend on $\mathbf{x}_1 - \mathbf{x}_2$, the additive structure of $\mathbb{R}^d$, which the order parameters lack.

EMBEDDED BOGOLIUBOV-BORN-GREEN-KIRKWOOD-YVON HIERARCHY

Embedding $\mathcal{M}$ in a vector space gives fast and slow coordinates for the order parameters too,

$$\underline{n} = \underline{\nu}_2 - \underline{\nu}_1, \quad \underline{N} = \frac{1}{2} (\underline{\nu}_2 + \underline{\nu}_1).$$

With $\underline{a} = \frac{1}{2} (\underline{B}(\underline{\nu}_1)^{-1} \underline{\varsigma}_1 + \underline{B}(\underline{\nu}_2)^{-1} \underline{\varsigma}_2)$, $\underline{b} = \underline{B}(\underline{\nu}_2)^{-1} \underline{\varsigma}_2 - \underline{B}(\underline{\nu}_1)^{-1} \underline{\varsigma}_1$ the f_2 equation becomes

$$\begin{aligned} & \frac{\partial f_2}{\partial t} + \frac{1}{2} \frac{\underline{p}_2 + \underline{p}_1}{m} \cdot \frac{\partial f_2}{\partial \underline{X}} + \underline{a} \cdot \frac{\partial f_2}{\partial \underline{N}} + \boxed{\underline{b} \cdot \frac{\partial f_2}{\partial \underline{n}}} + \boxed{\frac{\underline{p}_2 - \underline{p}_1}{m} \cdot \frac{\partial f_2}{\partial \underline{x}}} \\ & - \boxed{\frac{\partial \mathcal{W}(\underline{x}_1 - \underline{x}_2, \nu_1, \nu_2)}{\partial \underline{x}_1} \cdot \left(\frac{\partial f_2}{\partial \underline{p}_1} - \frac{\partial f_2}{\partial \underline{p}_2} \right)} - \boxed{\frac{\partial \mathcal{W}(\underline{x}_1 - \underline{x}_2, \nu_1, \nu_2)}{\partial \nu_1} \cdot \left(\frac{\partial f_2}{\partial \underline{\varsigma}_1} - \frac{\partial f_2}{\partial \underline{\varsigma}_2} \right)} = 0. \end{aligned}$$

VLASOV-BOLTZMANN EQUATION

4

VLASOV–TYPE EQUATION


Multiscale Model. Simul., in press, arXiv:2508.10744 (J. A. Carrillo, P. E. Farrell, A. Medaglia, U. Z.).

The separation of time scales gives the identity

$$\frac{\underline{p}_2 - \underline{p}_1}{m} \cdot \frac{\partial f_2}{\partial \mathbf{x}} = \frac{\partial \mathcal{W}}{\partial \mathbf{x}_1}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2) \cdot \left(\frac{\partial f_2}{\partial \underline{p}_1} - \frac{\partial f_2}{\partial \underline{p}_2} \right).$$

and substituting it into the first equation of the hierarchy gives

$$\begin{aligned} \frac{\partial f_1}{\partial t} + \frac{\underline{p}_1}{m} \cdot \frac{\partial f_1}{\partial \mathbf{x}_1} + \underline{B}(\nu_1)^{-1} \underline{\zeta}_1 \cdot \frac{\partial f_1}{\partial \nu_1} &= \int \frac{\underline{p}_2 - \underline{p}_1}{m} \cdot \frac{\partial f_2}{\partial \mathbf{x}} d\Gamma_2 \\ &+ \int \frac{\partial \mathcal{W}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2)}{\partial \nu_1} \cdot \frac{\partial f_2}{\partial \underline{\zeta}_1} d\Gamma_2. \end{aligned}$$

TWO SCALINGS IN ONE POTENTIAL

$$\mathcal{W}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2) = \mathcal{W}_c\left(\frac{\mathbf{x}_1 - \mathbf{x}_2}{d}, \nu_1, \nu_2\right) + \frac{1}{N} \mathcal{W}_{mf}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2).$$

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Steric part $\mathcal{W}_c$, Boltzmann–Grad scaling

Range d , strength c^2 , nd^2 fixed, $nd^3 \rightarrow 0$.

- ▶ Binary collisions at rate $1/\tau$, each with a new partner.
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Range $\ell \gg$ mean free path, strength $1/N$.

- ▶ N torques of size $1/N$ add up to an $\mathcal{O}(1)$ Vlasov force.
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In the mean-field integral, then, $\nabla_{\underline{z}_1} f_2 = f_1(\Gamma_2) \nabla_{\underline{z}_1} f_1(\Gamma_1) + o(1)$, while the orientations can stay correlated. This is the weak-order hypothesis, and both scalings survive in the limit.

WEAK-ORDER INTERACTIONS

The simplest closure assumes weak interactions,

$$f_2(\Gamma_1, \Gamma_2, t) = f_1(\Gamma_1, t)f_1(\Gamma_2, t).$$

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Weak-order interactions

We say that a system is governed by *weak-order interactions* if the derivatives of the two-particle distribution function with respect to the conjugate momenta factorise,

$$\nabla_{\underline{s}_i} f_2(\Gamma_i, \Gamma_j, t) = f_1(\Gamma_j, t) \nabla_{\underline{s}_i} f_1(\Gamma_i, t),$$

for $i \neq j$ and $i, j = 1, 2$. The order parameters themselves may stay correlated.

VLASOV–TYPE EQUATION

With weak-order interactions the first equation of the hierarchy closes,

$$\frac{\partial f}{\partial t} + \dot{\mathbf{x}} \cdot \nabla_{\mathbf{x}} f + \dot{\nu} \cdot \nabla_{\nu} f + \mathcal{V} \cdot \nabla_{\varsigma} f = C[f, f],$$

with the collision operator written through the transition probability W , and the Vlasov force $\mathcal{V}$,

$$\begin{aligned}
 C[f_1, f_1] = & \int d\Xi'_1 d\Xi'_2 d\Xi_2 \int_0^{\pi/2} \int_0^{2\pi} W(\Xi'_1, \Xi'_2 \mapsto \Xi_1, \Xi_2) f_1(\Gamma'_1, t) f_1(\Gamma'_2, t) \\
 & - W(\Xi_1, \Xi_2 \mapsto \Xi'_1, \Xi'_2) f_1(\Gamma_1, t) f_1(\Gamma_2, t) d\theta_2 d\varphi_2. \\
 \mathcal{V}(\mathbf{x}_1, \nu_1, t) = & \iiint \frac{\partial \mathcal{W}(\mathbf{x}_1 - \mathbf{x}_2, \nu_1, \nu_2)}{\partial \nu_1} f(\mathbf{x}_2, \nu_2, \varsigma_2, t) d\mathbf{x}_2 d\nu_2 d\varsigma_2.
 \end{aligned}$$

BOLTZMANN INEQUALITY AND THERMALISATION


J. Stat. Phys. 1981, 26, 795–801 (C. Cercignani, M. Lampis),
Multiscale Model. Simul., in press, arXiv:2508.10744 (J. A. Carrillo, P. E. Farrell, A. Medaglia, U. Z.).

The collision operator satisfies the Boltzmann inequality (Theorem 6.2), so the system thermalises,

$$\int d\Xi \log(f(\Gamma, t)) C[f, f] \leq 0.$$

Theorem 6.3 (Maxwellian)

For a transitive action, the only distributions annihilating the collision operator are

$$\bar{f}(\Gamma, t) = \exp \left(c + \underline{a} \cdot \underline{p} + A_\nu \underline{b} \cdot \underline{\varsigma} + d(m^{-1} \underline{p} \cdot \underline{p} + \underline{\varsigma} \cdot \underline{B}(\nu)^{-1} \underline{\varsigma}) \right),$$

with $c \in \mathbb{R}$, $d < 0$, $\underline{a} \in \mathbb{R}^d$ and $\underline{b} \in \mathfrak{so}(d)$.

DIRECT SIMULATION MONTE CARLO

5

THE SPATIALLY HOMOGENEOUS REGIME



arXiv 2026, 2609.31072 (J. A. Carrillo, P. E. Farrell, A. Medaglia, U. Z.).

Without the position, and with the Maxwellian kernel (all pairs equally likely to collide),

$$\frac{\partial f}{\partial t} + \underline{\underline{B}}(\nu)^{-1} \underline{\underline{\zeta}} \cdot \nabla_{\nu} f + \nu \cdot \nabla_{\underline{\underline{\zeta}}} f = \frac{1}{\tau} \iiint (f' f'_2 - f f_2) d\underline{\underline{p}}_2 d\nu_2 d\underline{\underline{\zeta}}_2.$$

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For calamitic molecules in two dimensions, $\mathcal{M} = \mathbb{S}^1 \ni \theta$ and $\varsigma = b\omega$, with $\mathcal{V}$ now the angular acceleration,

$$\partial_t f + \omega \partial_{\theta} f + \mathcal{V} \partial_{\omega} f = \frac{1}{\tau} \iiint (f' f'_2 - f f_2) d\underline{p}_2 d\theta_2 d\omega_2.$$

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- ▶ A tensor grid with n points per direction costs n^d on the phase space $(\underline{p}, \nu, \underline{\varsigma})$.
- ▶ DSMC represents f by particles and converges at the Monte Carlo rate, which does not depend on the dimension.

OPERATOR SPLITTING

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We use the symmetric Strang splitting

$$f^{n+1} = \mathcal{T}_{\Delta t/2} \left(\mathcal{Q}_{\Delta t} \left(\mathcal{T}_{\Delta t/2}(f^n) \right) \right).$$

The collision step is first order, so the scheme is not second order. The symmetric form is kept for the energy balance.

TRANSPORT STEP: SYMPLECTIC INTEGRATION

1. Particles

$$f^N = \frac{1}{N} \sum_i \delta(\underline{p} - \underline{p}_i) \otimes \delta(\nu - \nu_i) \otimes \delta(\underline{\varsigma} - \underline{\varsigma}_i).$$

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3. Störmer–Verlet

drift $\mathcal{D}_\delta : \theta_i \mapsto \theta_i + \omega_i \delta$ (geodesic flow),

kick $\mathcal{K}_\delta : \underline{\varsigma}_i \mapsto \underline{\varsigma}_i + \mathcal{V}(\nu_i) \delta$,

$$\mathcal{T}_{\Delta t} \approx \mathcal{D}_{\Delta t/2} \mathcal{K}_{\Delta t} \mathcal{D}_{\Delta t/2}.$$

TRANSPORT STEP: SYMPLECTIC INTEGRATION

1. Particles

$$f^N = \frac{1}{N} \sum_i \delta(\underline{p} - \underline{p}_i) \otimes \delta(\nu - \nu_i) \otimes \delta(\underline{\varsigma} - \underline{\varsigma}_i).$$

2. Characteristics

$$\frac{d\nu_i}{dt} = \underline{B}(\nu_i)^{-1} \underline{\varsigma}_i, \quad \frac{d\underline{\varsigma}_i}{dt} = \mathcal{V}(\nu_i).$$

3. Störmer–Verlet

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Proposition 7.1 (Conservation of the total energy)

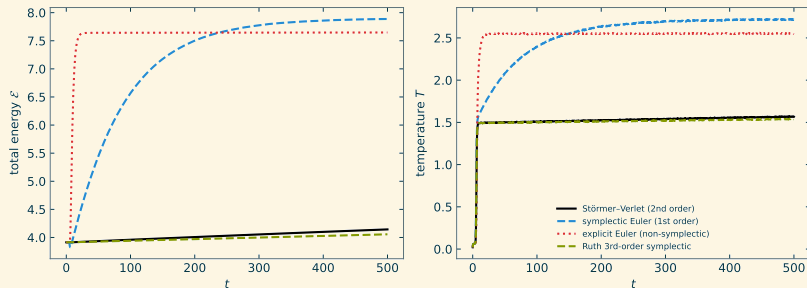
For a transitive action, elastic collisions and a conservative force $\mathcal{V} = -\nabla_\nu(\mathcal{W} * \lambda)$ with symmetric $\mathcal{W}$, the total energy

$$\mathcal{E}_{\text{tot}} = \int \frac{1}{2} \left(\frac{|\underline{p}|^2}{m} + \underline{\varsigma} \cdot \underline{B}^{-1} \underline{\varsigma} \right) f + \frac{1}{2} \iint \mathcal{W} \lambda \lambda$$

is conserved.

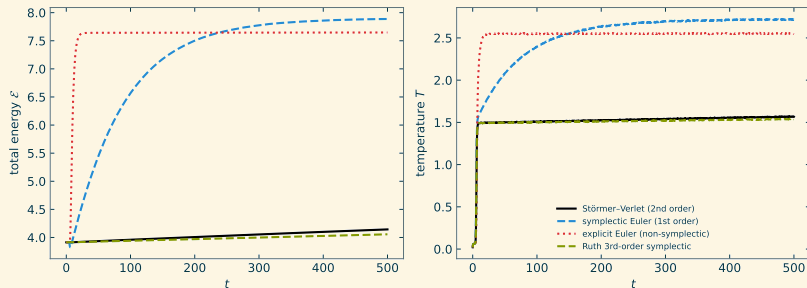
Ordering turns interaction energy into kinetic energy, and the scheme has to track that exchange.

WHY A SECOND-ORDER INTEGRATOR



Hard-needle run integrated with explicit Euler, symplectic Euler, Störmer–Verlet and the third-order Ruth scheme, total energy (left) and temperature (right).

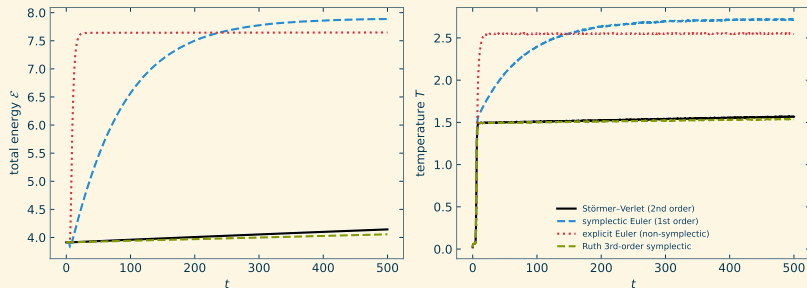
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- ▶ Both first-order schemes drift by a factor of two and heat the system spuriously. The kick breaks symplecticity, so the order of the scheme is what controls the error.
- ▶ Störmer–Verlet holds the plateau within a few percent, and Ruth improves on it only marginally.

COLLISION STEP: THREE STRATEGIES

In each strategy the explicit Euler step is a convex combination, so it can be carried out by randomly resampling particles.

BGK resampling

$$f^{n+1} = \left(1 - \frac{\Delta t}{\tau}\right) f^n + \frac{\Delta t}{\tau} \bar{f}[f^n].$$

- ▶ Redraw $(\underline{p}, \underline{\varsigma})$ from the local Maxwellian.
- ▶ Held at T_{bath} , this is the Andersen thermostat.

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- ▶ $\text{Sround}(N\Delta t/2\tau)$ random pairs collide.

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Bird's method

$$\begin{aligned} \partial_t f &= \iiint W f' f'_2 \\ &+ \iiint (w_{\max} - W) f f_2 \\ &- w_{\max} f. \end{aligned}$$

- ▶ Null collisions handle any kernel $W \leq w_{\max}$.
- ▶ Accept with probability $W/w_{\max}$.

THE BINARY COLLISION OF ROD-LIKE MOLECULES

For two-dimensional rods the impulse of the collision response model is

$$J = - \frac{\underline{V} \cdot \underline{n}}{\frac{2}{m} + \left[\mathbb{I}_1^\dagger(\underline{r}_1 \times \underline{n}) \times \underline{r}_1 + \mathbb{I}_2^\dagger(\underline{r}_2 \times \underline{n}) \times \underline{r}_2 \right] \cdot \underline{n}},$$

with $\mathbb{I}^\dagger$ the pseudo-inverse of the inertia tensor, and the post-collisional states are

$$\begin{aligned} \dot{\underline{x}}'_1 &= \dot{\underline{x}}_1 - (1 + e_v) \frac{J}{m} \underline{n}, & \dot{\underline{x}}'_2 &= \dot{\underline{x}}_2 + (1 + e_v) \frac{J}{m} \underline{n}, \\ \underline{\omega}'_1 &= \underline{\omega}_1 - (1 + e_\omega) J \mathbb{I}_1^\dagger(\underline{r}_1 \times \underline{n}), & \underline{\omega}'_2 &= \underline{\omega}_2 + (1 + e_\omega) J \mathbb{I}_2^\dagger(\underline{r}_2 \times \underline{n}). \end{aligned}$$

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- ▶ In DSMC we sample the contact normal $\underline{n} = (\cos \varphi, \sin \varphi)$, $\varphi \sim \mathcal{U}([0, 2\pi])$, and the arm $\underline{r}_1 = \ell \underline{\nu}_1$, $\ell \sim \mathcal{U}([0, 1])$.
- ▶ With $e_v = e_\omega = 1$ the collision is elastic. It conserves linear momentum, angular momentum and energy, and leaves θ unchanged.

MEAN FIELD, ORDER AND PARAMETERS

To compute the mean-field force,

- ▶ phenomenological potentials need only the circular mean, $\hat{\theta} = \arg \sum_i e^{i\theta_i}$.
- ▶ a general $\mathcal{W}$ needs λ itself, estimated by cloud-in-cell on N_θ cells, after which $\Phi = \mathcal{W} * \lambda$ costs $\mathcal{O}(N_\theta^2)$.

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For axes the first moment vanishes, so we measure order by

$$\sigma^2(\theta) = 1 - \left| \frac{1}{N} \sum_{i=1}^N e^{2i\theta_i} \right|,$$

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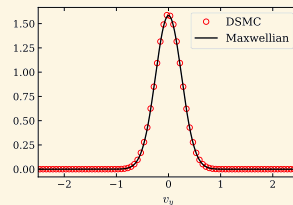
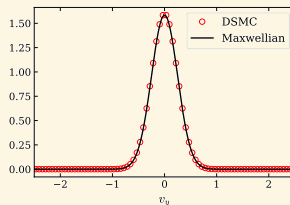
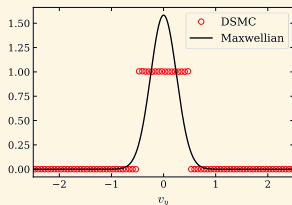
which is 1 when isotropic and tends to 0 under alignment.

Parameters of all the experiments

- ▶ $N = 10^7$ particles, $N_\theta = 256$ cells.
- ▶ $m = 1$, $L = \sqrt{12}$, so that $b = mL^2/12 = 1$.
- ▶ Elastic collisions, Maxwellian kernel unless stated.
- ▶ $\Delta t = 0.05$, $t_f = 50$.
- ▶ $1/\tau = 10$ for thermalisation, $1/\tau = 4$ with a mean field.

THERMALISATION AND THE H-THEOREM

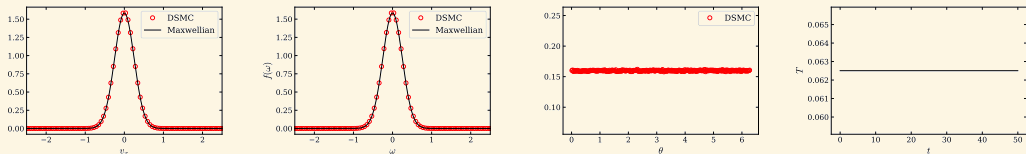
We set $\mathcal{W} = 0$ and start from a datum that is bimodal in p_y , Gaussian in p_x and ω , and uniform in θ .



The v_y marginal at $t = 0, 5$ and 50 relaxes from the bimodal datum to the Maxwellian (solid line).

THERMALISATION AND THE H-THEOREM

We set $\mathcal{W} = 0$ and start from a datum that is bimodal in p_y , Gaussian in p_x and ω , and uniform in θ .



The v_x , ω and θ marginals at $t = 50$, and the temperature history. The orientational marginal stays uniform, since the collisions do not change the orientations.

This is the H-theorem seen numerically. The relaxation is monotone and ends at the Maxwellian.

MEAN-FIELD ALIGNMENT: A LINEAR RESTORING FORCE

With $\mathcal{W}(\theta, \omega) = \frac{\alpha}{2}|\theta - \hat{\theta}|^2 + \beta\omega\theta$, $\hat{\theta}$ the circular mean, the force is $\mathcal{V} = -\alpha(\theta - \hat{\theta}) - \beta\omega$ and the characteristics are linear,

$$\frac{d}{dt} \begin{bmatrix} \theta_i \\ \omega_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\alpha & -\beta \end{bmatrix} \begin{bmatrix} \theta_i \\ \omega_i \end{bmatrix} + \alpha \begin{bmatrix} 0 \\ \hat{\theta} \end{bmatrix},$$

with eigenvalues $\mu_{1,2} = \frac{1}{2}(-\beta \pm \sqrt{\beta^2 - 4\alpha})$.

- ▶ $\beta = 0, \alpha > 0$: a centre.
- ▶ $\beta > 0, \alpha > \beta^2/4$: a stable spiral.
- ▶ $\beta > 0, 0 < \alpha < \beta^2/4$: a stable node.
- ▶ $\alpha < 0$: a saddle, no alignment.

The Kuramoto force $\mathcal{V} = -\sin(\theta - \hat{\theta})$ is its nonlinear version, the torque of an external field.

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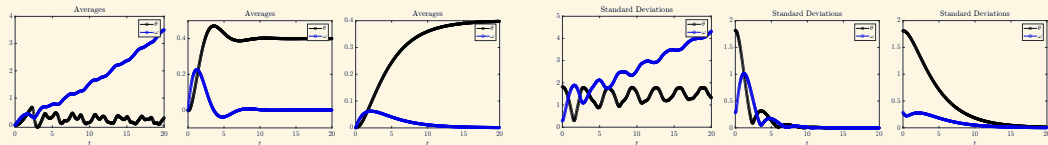
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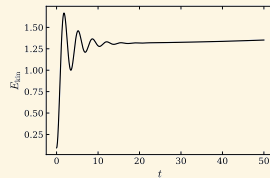
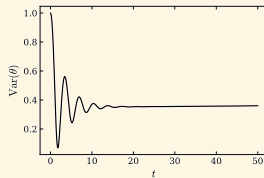
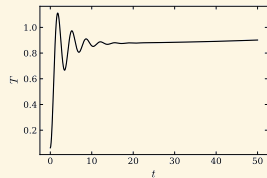
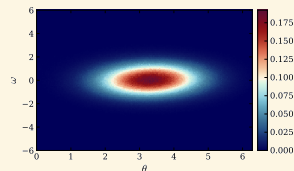
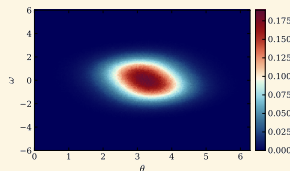
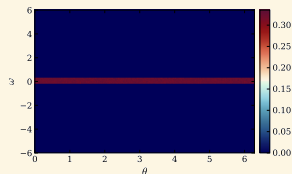
The Kuramoto force $\mathcal{V} = -\sin(\theta - \hat{\theta})$ is its nonlinear version, the torque of an external field.



Mean (left three) and standard deviation (right three) of the particle ODEs: centre ($\beta = 0, \alpha = 1$), spiral ($\beta = 1, \alpha = 1$), node ($\beta = 1, \alpha = 0.25$).

MEAN-FIELD ALIGNMENT: DSMC

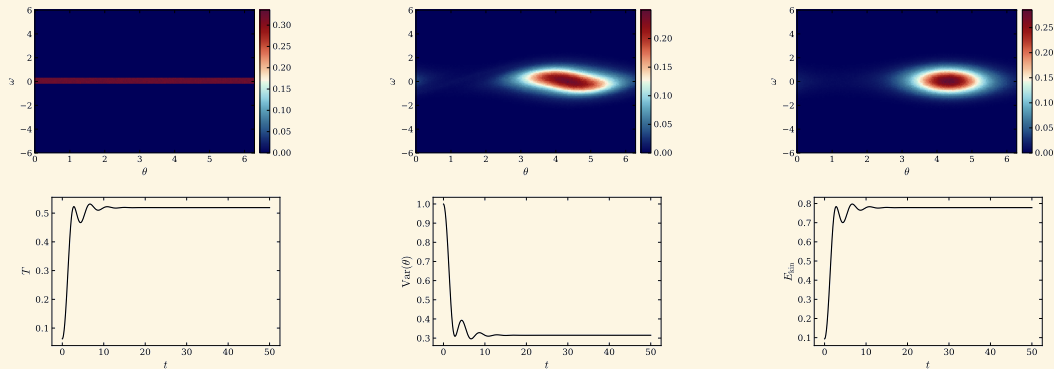
Quadratic potential, $\beta = 0$. Top: (θ, ω) density at $t = 0, 5, 10$. Bottom: temperature, circular variance, kinetic energy.



The rods align, and the collisions damp the oscillations of the collisionless ODEs.

MEAN-FIELD ALIGNMENT: DSMC

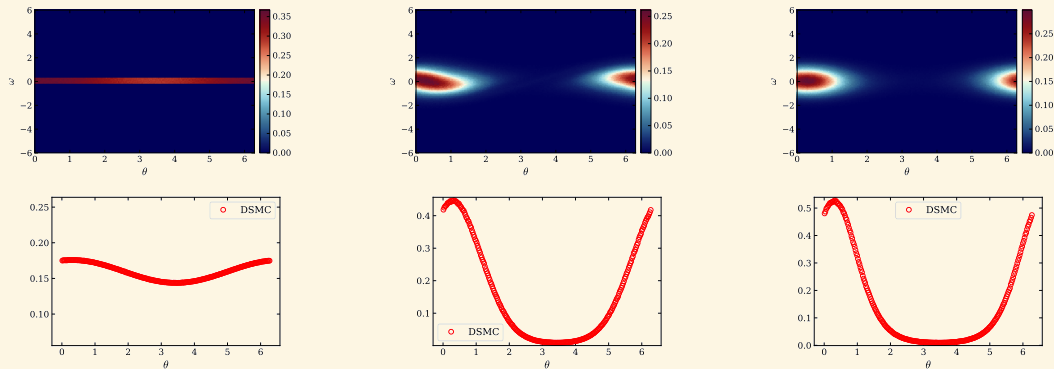
Kuramoto potential. Top: (θ, ω) density at $t = 0, 5, 10$. Bottom: temperature, circular variance, kinetic energy.



The quadratic force grows with the separation, so it condenses the rods faster.

NUCLEATION OF ORDER

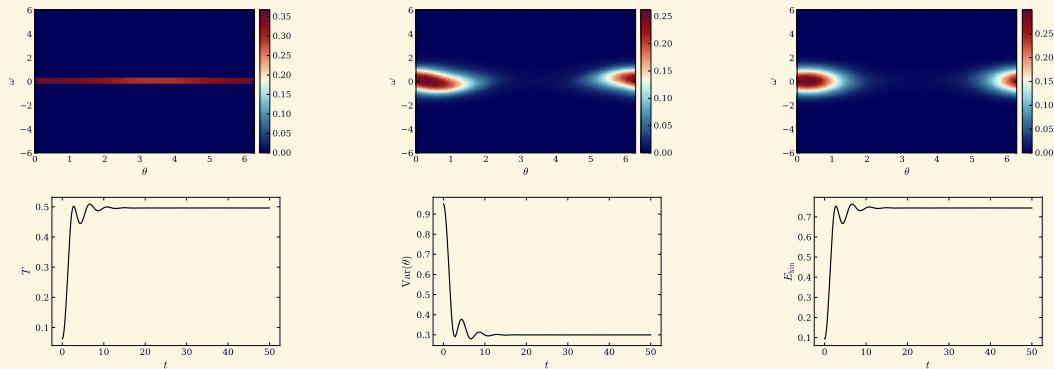
Kuramoto potential, perturbed isotropic datum. Top: (θ, ω) density at $t = 0, 5, 10$. Bottom: θ marginals.



The anisotropy starts below the statistical noise, but grows exponentially.

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FORMATION OF POLAR ORDER AT EQUILIBRIUM

Collisions thermalise only the momenta, so we take the Gibbs ansatz

$$\lambda(\underline{\nu}) = \frac{1}{Z} \exp \left(- \frac{1}{k_B T} \int_{\mathbb{S}^2} \mathcal{W}(\underline{\nu}, \underline{\nu}_*) \lambda(\underline{\nu}_*) d\underline{\nu}_* \right).$$

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Proposition 8.1 (Kuramoto-type potentials)

For $\mathcal{W}_K = -\underline{\nu} \cdot \underline{\nu}_*$ the isotropic state always solves the Gibbs equation, and $\underline{\gamma} = \int \underline{\nu} \lambda = s \hat{\underline{\nu}}$ with

$$s = I(s) = \coth \left(\frac{s}{k_B T} \right) - \frac{k_B T}{s}.$$

The isotropic branch has a pitchfork at $k_B T = 1/3$, below which a unique $s^* > 0$ appears.

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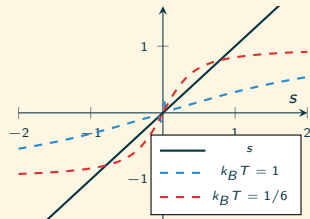
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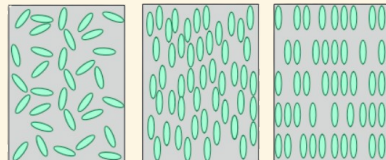
The Langevin function $I(s)$ against s . Above $k_B T = 1/3$ only the isotropic root exists. Below it, two signed roots $\pm s^*$ describe the same aligned state.

ONSAGER STERIC POTENTIAL

We now drive the ordering by the excluded area of two thin rods of length L ,

$$\mathcal{W}_{\text{Ons}}(\theta, \theta_*) = L^2 |\sin(\theta - \theta_*)|,$$

minimal for parallel and anti-parallel rods, so it favours alignment along a common director.



Liquid phase:
no orientation
or periodicity

Nematic phase:
orientation, no
periodicity

Smectic phase:
orientation with
some periodicity

Decreasing temperature $\rightarrow T$

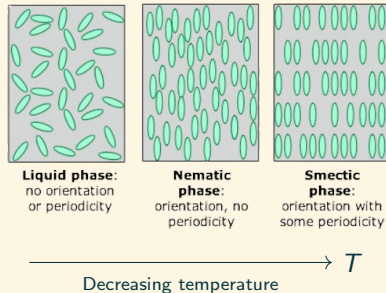
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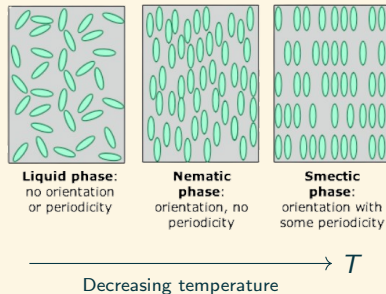
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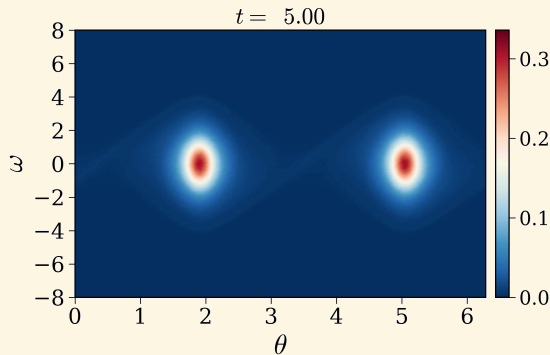
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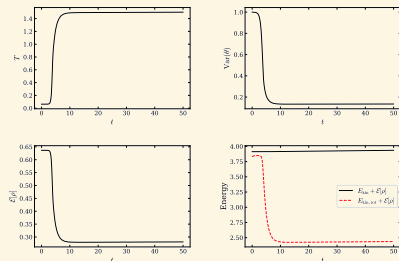
- ▶ It is invariant under $\theta \mapsto \theta + \pi$, so the kernel carries the head–tail symmetry, as in Onsager's theory.
- ▶ The full mean-field force $\mathcal{V} = -\partial_\theta(\mathcal{W}_{\text{Ons}} * \lambda)$ is computed by cloud-in-cell.
- ▶ There is no thermostat, so $\mathcal{E}_{\text{tot}}$ is conserved (Proposition 7.1).



ONSAGER POTENTIAL: MICROCANONICAL RUN



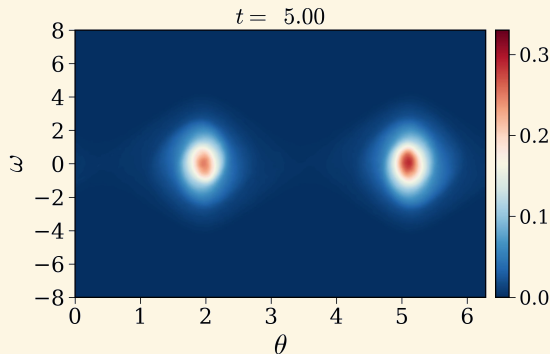
The (θ, ω) density for $t \in [0, 50]$ (click to play).



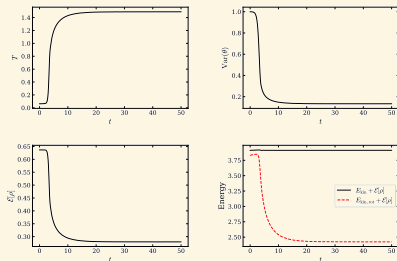
Temperature, circular variance, interaction and total energy.

- ▶ Two head–tail symmetric bumps form.
- ▶ Interaction energy turns into kinetic energy. The total energy drifts by about 0.6%.

ONSAGER POTENTIAL: HARD-NEEDLE KERNEL



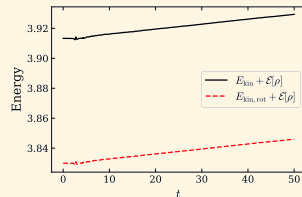
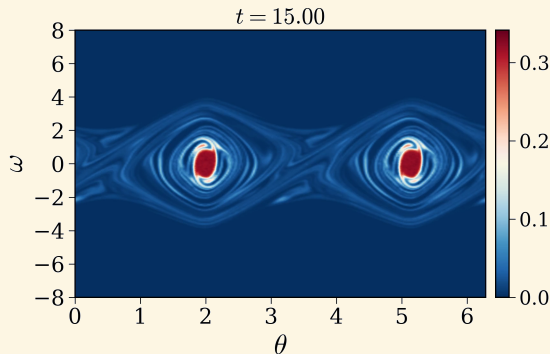
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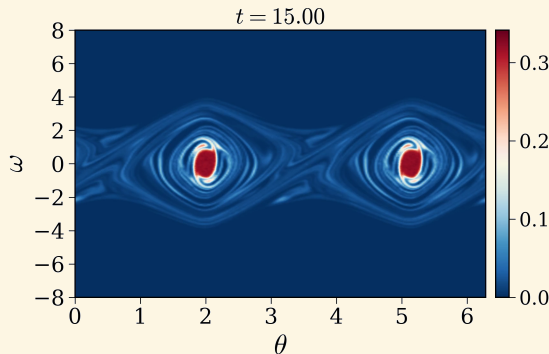
- ▶ The physical kernel gives the same ordering.
- ▶ The total energy fluctuates by about 0.2%.

COLLISIONLESS DYNAMICS: FILAMENTATION

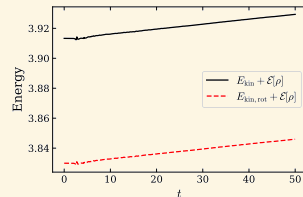


Onsager potential with the collisions switched off: the (θ, ω) density for $t \in [0, 50]$, click to play.

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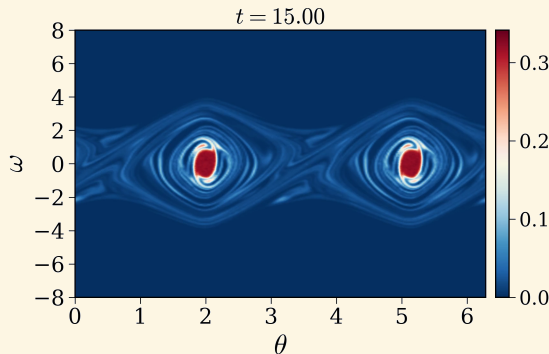


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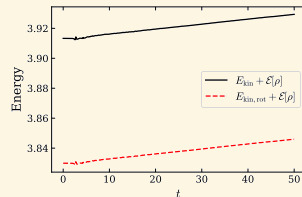


- The band rolls up into vortices and ever finer filaments, as in the diocotron instability of non-neutral plasmas.

COLLISIONLESS DYNAMICS: FILAMENTATION



Onsager potential with the collisions switched off: the (θ, ω) density for $t \in [0, 50]$, click to play.



- ▶ The band rolls up into vortices and ever finer filaments, as in the diocotron instability of non-neutral plasmas.
- ▶ Coarse observables converge by phase mixing, while the density never relaxes pointwise. The energy drifts by about 0.4%.

GRADIENT-FLOW STRUCTURE OF THE MARGINAL DYNAMICS

6

FROM THE KINETIC EQUATION TO A DARCY LAW

In the spatially homogeneous setting, with the BGK operator as collision term,

$$\partial_t f + \underline{\underline{B}}(\nu)^{-1} \underline{\underline{\zeta}} \cdot \nabla_\nu f + \mathcal{V} \cdot \nabla_{\underline{\underline{\zeta}}} f = \frac{1}{\tau} (\bar{f} - f), \quad \mathcal{V} = -\nabla_\nu (\mathcal{W} * \lambda).$$

Integrating over the momenta gives a continuity equation for the orientations,

$\partial_t \lambda + \nabla_\nu \cdot \mu = 0$, but the flux $\mu = \int \underline{\underline{B}}^{-1} \underline{\underline{\zeta}} f$ still depends on f .

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When collisions are fast, f stays close to the Maxwellian. For a transitive action with $\underline{\underline{B}} = b \underline{\underline{I}}$ the flux then closes (Lemma 11.1), and after rescaling time by τ/b

$$\mu = -\lambda \nabla_\nu p_E, \quad p_E = \beta \log \lambda + \mathcal{W} * \lambda, \quad \beta = k_B T.$$

A GRADIENT FLOW OF THE FREE ENERGY

The pressure p_E is, up to a constant, the variational derivative of

$$E[\lambda] = \beta \int_{\mathcal{M}} \lambda \log \lambda + \frac{1}{2} \iint \mathcal{W}(\nu, \nu_*) \lambda(\nu) \lambda(\nu_*),$$

so the marginal equation is an aggregation–diffusion equation along which E decreases,

$$\partial_t \lambda = \nabla_\nu \cdot \left(\lambda \nabla_\nu \frac{\delta E}{\delta \lambda} \right), \quad \frac{d}{dt} E[\lambda] = - \int_{\mathcal{M}} \lambda \left| \nabla_\nu \frac{\delta E}{\delta \lambda} \right|^2 \leq 0.$$



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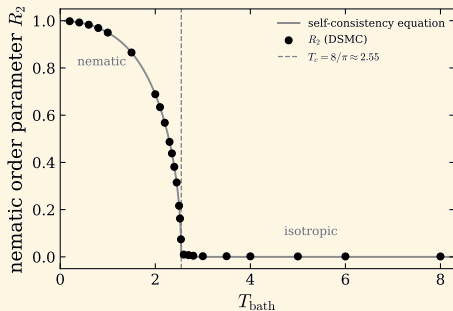
Proposition 11.1

The stationary states are the solutions of the Gibbs equation $\lambda_\infty = Z^{-1} \exp(-\beta^{-1} \mathcal{W} * \lambda_\infty)$, that is the equilibria of Chapters 8 and 9.



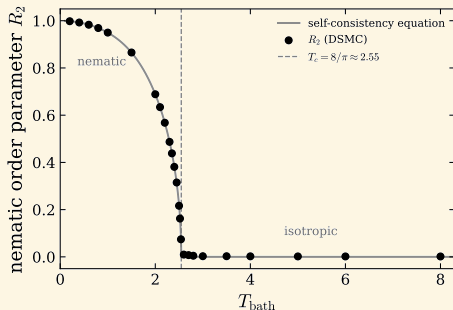
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WHAT THE THERMOSTATTED DSMC SHOWS

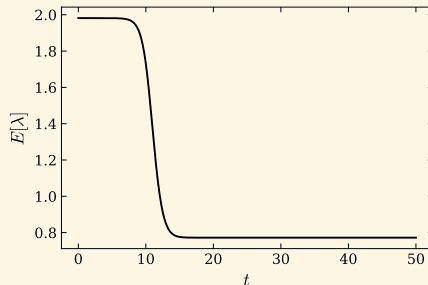


Equilibria for $L^2 = 12$. The order vanishes at $T_c = 2L^2/(3\pi) = 8/\pi$, the threshold of Proposition 9.2.

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Dynamics at $T_{\text{bath}} = 1$. The free energy falls from 1.98 to 0.77, its value at the Gibbs state.

ZERO TEMPERATURE IN QUANTILE VARIABLES

At $\beta = 0$, on $\mathbb{RP}^1$ with the Onsager kernel $\mathcal{W} = L^2|\sin(\theta - \theta_*)|$, only the aggregation is left.



New Trends in Math. Phys. 2005 (J. A. Carrillo, G. Toscani), *in preparation* (J. A. Carrillo, U. Z.).

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Writing λ through its quantile function G_λ , measures supported in an arc of length at most $\pi/2$ satisfy $W_2(\mu, \nu) = \|G_\nu - G_\mu\|_{L^2(0,1)}$ (Proposition 11.2). The gradient flow then becomes an ODE in $L^2(0, 1)$ (Proposition 11.3),

$$\partial_t G(p, t) = -L^2 \int_0^1 \cos(G(p, t) - G(p_*, t)) \operatorname{sgn}(p - p_*) dp_*.$$



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The ODE is well posed (Lemma 11.2), and if $\lambda^0 \leq \Lambda$ the quantiles stay ordered at least until $t = 1/(2L^2\Lambda)$ (Lemma 11.3).



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FINITE-TIME CONCENTRATION

Proposition 11.4

If the initial orientations lie in an arc of length $D(0) < \pi/2$, the diameter of the support satisfies

$$D(0) - 2L^2 t \leq D(t) \leq D(0) - 2L^2 \cos(D(0)) t,$$

so λ collapses onto a single director in finite time.

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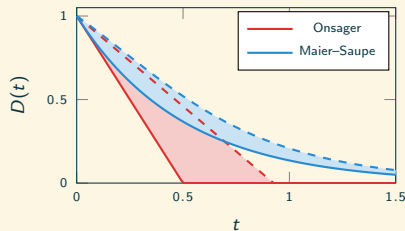
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Bounds on the diameter for $D(0) = 1$ and $L^2 = 1$.

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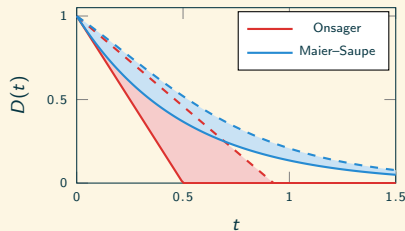
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The Onsager kernel has a corner at alignment, so the attraction between nearly aligned rods stays of order L^2 . For Maier–Saupe it vanishes linearly with the angle.



Bounds on the diameter for $D(0) = 1$ and $L^2 = 1$.

OUTLOOK

7

SUMMARY AND OUTLOOK

In this thesis

- ▶ The order parameter manifold as a unified description of microscopic order.
- ▶ A BBGKY hierarchy on the enlarged phase space, and a Vlasov–Boltzmann equation with its H-theorem.
- ▶ A DSMC method, with the calamitic and nematic case studies.
- ▶ A BGK model operator, and the gradient-flow structure of the orientational dynamics, with finite-time concentration on $\mathbb{RP}^1$.

Open directions

- ▶ Gradient flows beyond one dimension: the torus first, then $\mathbb{S}^2$ and $\mathbb{RP}^2$, where curvature enters.
- ▶ A rigorous overdamped limit, via relative entropy methods.
- ▶ The flow past the first collision.
- ▶ Hydrodynamic limits at a generic degree of order, and the grazing limit of the collision operator.

THANK YOU!

Kinetic Modelling of Ordered Fluids

UMBERTO ZERBINATI